Daniel Heinz

Head of the Digital Service Design & Innovation Lab
Karlsruhe Institute of Technology

Curriculum vitae · November 2026

daniel.heinz@kit.edu · Institutional profile · Read this CV online

Research

I study how work gets organized when AI agents act in firms, in new ventures, and between organizations, and I design, build, and evaluate AI systems in the field.

Observe: qualitative research (interviews, case studies, taxonomies) and computationally intensive theory construction. Build: design science and deployment in organizations and ventures. Test: field experiments and interviews.

Academic appointments

Since 2024 · Head of the Digital Service Design & Innovation Lab
Karlsruhe Institute of Technology

2021–2024 · Research Associate
Karlsruhe Institute of Technology

Education

2025 · Dr. rer. pol. in Information Systems
Karlsruhe Institute of Technology
Dissertation: Smart Service Innovation in Incumbent Firms. Summa cum laude.

2021 · M.Sc. in Industrial Engineering & Management
Karlsruhe Institute of Technology
With distinction.

2018 · B.Sc. in Industrial Engineering & Management
Karlsruhe Institute of Technology

Visiting research stays at the University of Cambridge (2024) and The University of Queensland (2022).

Academic service

President of AIS SIG Services
Co-head of the Conferences Division, Die WI e.V.
Chair for Workshops, Tutorials & Community Events, WI 2027

Volunteering

Outside work I volunteer with DPSG, Germany’s largest scout association. I lead its program for 15- to 21-year-olds in the Freiburg diocese, train youth leaders, and from 2023 to 2025 led CrOver, an international camp for about 250 scouts supported by €99,495 in Erasmus+ funding.

Selected funded projects

I work on publicly funded projects and industry collaborations at KIT, with responsibilities ranging from project leadership to proposal development.

SHAPE · 2024–2027
BMBF / BMFTR · DigiLeistDAT

How can service ecosystems coordinate people, expertise, and resources to develop digital services?
Co-wrote the proposal for the KIT sub-project; sub-project lead

ZirkulEA · 2022–2025
BMBF · MobilKreis

How can shared lifecycle data support circular value chains for electric powertrains?
Co-wrote the proposal for the KIT sub-project; sub-project lead

bi.smart · 2021–2024
BMBF · Advanced Systems Engineering

How can small and medium-sized manufacturers develop connected products and digital services around customer needs?
Project lead

AI Adoption · Since 2025
Industry-funded collaboration · BASF

How do organizations move from AI pilots to sustained use across the enterprise?
Shaped the research agenda of the proposal; project lead

Truck Traceability · 2025–2026
Industry-funded collaboration · Daimler Truck

How can lifecycle data help companies reuse, remanufacture, and recycle trucks and their components?
Led the acquisition; project lead

Doctoral research

I provide day-to-day research advising for Linda Sagnier Eckert, Jonas Liebschner, and Ida Meier. Prof. Gerhard Satzger provides formal doctoral supervision for these three researchers. I also provide day-to-day research supervision for Pandora Dietle, whose formal doctoral supervisor is Prof. Maximilian Förster.

Linda Sagnier Eckert · Since April 2024
Traceability for the circular economy

Jonas Liebschner · Since April 2024
AI assistants for innovation and entrepreneurship

Ida Meier · Since September 2025
AI adoption in organizations

Pandora Dietle · Since November 2026
Human–agentic AI collaboration

Journal articles

A Mechanism-Based Theory of Legacy Enterprise Information System Reconfiguration in Incumbent Firms
Daniel Heinz & Christoph F. Breidbach
Journal of Information Technology 41(3), 221–249 · 2026
https://doi.org/10.1177/02683962261461296

Smart Services in B2B Contexts: A Taxonomy and Archetypes
Daniel Heinz, Marcel Fassnacht, Carina Benz, Philipp Luchner & Gerhard Satzger
Information & Management 63(7), Article 104390 · 2026
https://doi.org/10.1016/j.im.2026.104390

Backcasting the Future of the Automotive Industry: Leveraging Digital Technologies for the Circular Economy
Linda Sagnier Eckert, Ann-Sophie Finner, Daniel Heinz & Gerhard Satzger
Communications of the Association for Information Systems 58(1), 878–912 · 2026 · In press
https://doi.org/10.17705/1CAIS.05835

Selecting Data Assets in Data Marketplaces: Leveraging Machine Learning and Explainable AI for Value Quantification
Dominik Martin, Daniel Heinz, Moritz Glauner & Niklas Kühl
Business & Information Systems Engineering 68(3), 523–558 · 2026
https://doi.org/10.1007/s12599-025-00940-8

Conceptualizing Hybrid Intelligent Service Ecosystems
Christian Bartelheimer, Daniel Heinz, Sarah Hönigsberg, Dominik Siemon, Mahei Manhai Li, Timo Strohmann, Jens Poeppelbuss & Christoph Peters
Electronic Markets 35(1), Article 63 · 2025
https://doi.org/10.1007/s12525-025-00798-4

Unlocking the Value from Car Data: A Taxonomy and Archetypes of Connected Car Business Models
Felix Sterk, Alexander Stocker, Daniel Heinz & Christof Weinhardt
Electronic Markets 34(1), Article 13 · 2024
https://doi.org/10.1007/s12525-024-00692-5

Data Sharing Practices: The Interplay of Data, Organizational Structures, and Network Dynamics
Marcel Fassnacht, Jannis Leimstoll, Carina Benz, Daniel Heinz & Gerhard Satzger
Electronic Markets 34(1), Article 47 · 2024
https://doi.org/10.1007/s12525-024-00732-0

Creating Customer Value from Data: Foundations and Archetypes of Analytics-Based Services
Fabian Hunke, Daniel Heinz & Gerhard Satzger
Electronic Markets 32(2), 503–521 · 2022
https://doi.org/10.1007/s12525-021-00506-y

Exploring the Potential of Smart Service Systems: A Multi-Actor View on Affordances and Their Actualization
Daniel Heinz, Carina Benz, Jan Bode, Fabian Hunke & Gerhard Satzger
Journal of Service Management Research 6(2), 132–146 · 2022
https://doi.org/10.5771/2511-8676-2022-2-132

Selected conference papers

From Pilot to Production: Microfoundations and Responsibility Patterns for Enterprise AI Projects
Daniel Hendriks, Ida Meier, Henrik Wiegand, Carina Benz, Daniel Heinz, and Philipp Spitzer
ICIS · 2026 · Accepted
https://doi.org/10.5445/IR/1000197565

The Adoption of AI-Based Assistants in Startup Innovation: Configuring Human-AI Agency in Entrepreneurial Work
Jonas Liebschner, Ida Meier, Daniel Heinz, and Gerhard Satzger
ECIS · 2026
https://aisel.aisnet.org/ecis2026/is_adopt/is_adopt/20/

Designing AI-Based Entrepreneurial Coaching Systems
Jonas Liebschner, Daniel Heinz, and Gerhard Satzger
DESRIST · 2026
https://doi.org/10.1007/978-3-032-28313-9_2

Making the Implicit Explicit: A Human-in-the-Loop AI Pipeline for Excavating and Making Use of Latent Design Knowledge
Timo Strohmann, Linda Sagnier Eckert, Daniel Heinz, Thorsten Schoormann, Christoph Hoppe-Ludwig, and Anne Ixmeier
DESRIST · 2026
https://doi.org/10.1007/978-3-032-28316-0_3

Reallocating Uncertainty in Incumbent Firms through Digital Platforms: The Case of Google's Automotive Ecosystem Involvement
Felix Sterk, Daniel Heinz, Petra Hengstler, and Christof Weinhardt
ICIS · 2023
https://aisel.aisnet.org/icis2023/gov_strategy/gov_strategy/3/

Conceptualizing Digital Resilience for AI-Based Information Systems
Max Schemmer, Daniel Heinz, Lucas Baier, Michael Vössing, and Niklas Kühl
ECIS (Research-in-Progress) · 2021
https://aisel.aisnet.org/ecis2021_rip/44/

Teaching and advising

Digital Services: Innovation & Business Models
Master's course · KIT

Building Digital Services with AI
MSc practical seminar · KIT · Winter 2026/27

Philosophy of Science for PhD Students
Doctoral seminar · KIT · Winter 2026/27

Qualitative Research (incl. case study)
Online module · Research Training@WIN · Module lead · Winter 2026/27

Digital Services at HECTOR School
Executive education · KIT

I provide day-to-day doctoral advising and supervise bachelor's and master's theses.